



Starbucks

Starbucks Coffee

Welcome to Starbucks, your ultimate destination for exceptional coffee experiences. At Starbucks, we believe in crafting moments of joy and connection through every cup of coffee we serve.





Our Team

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PRIYA



SANJANA

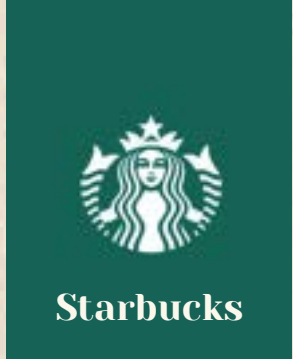


VARUNA



ASHRITHA

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Starbucks Ordering Patterns

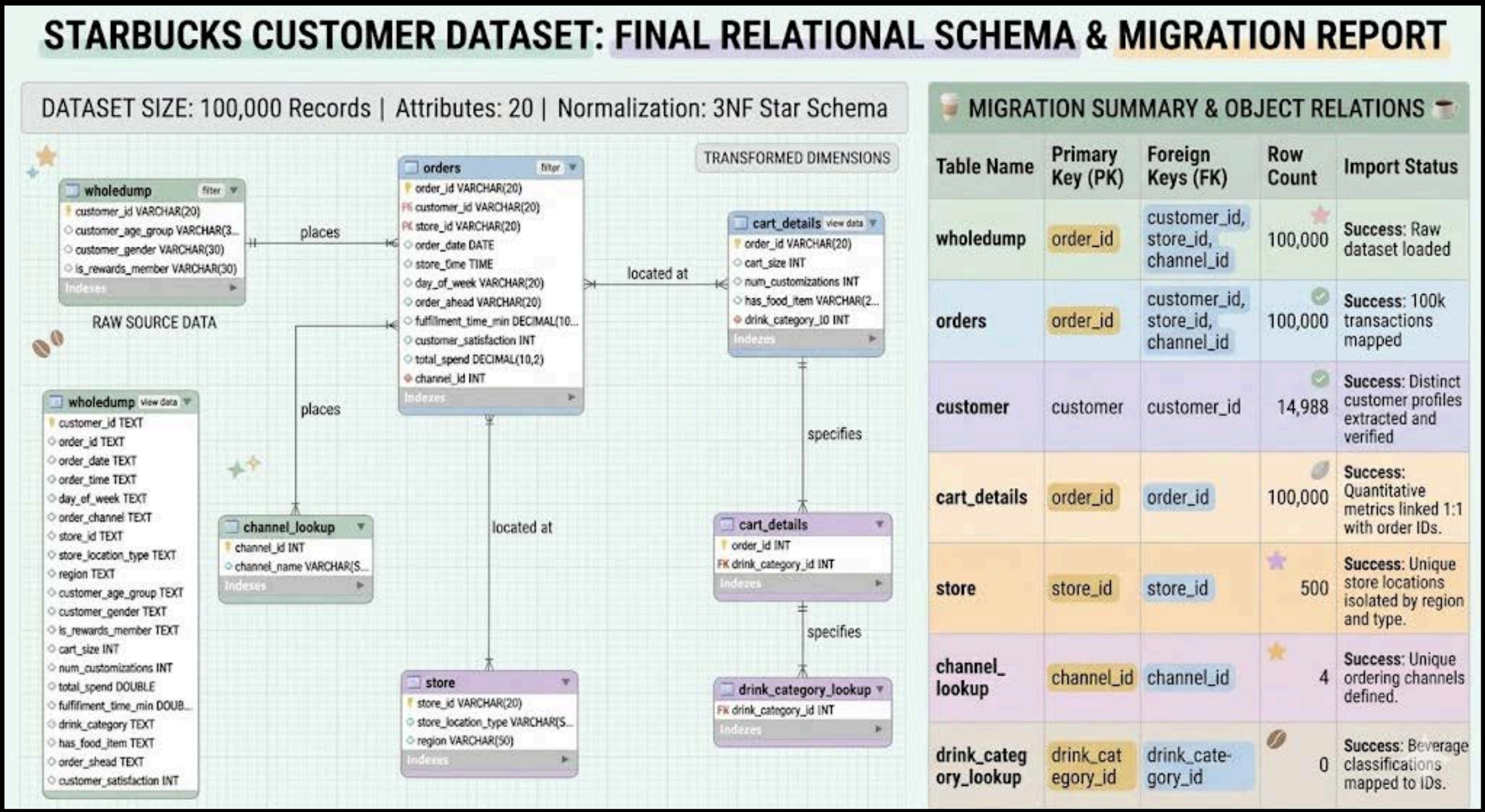


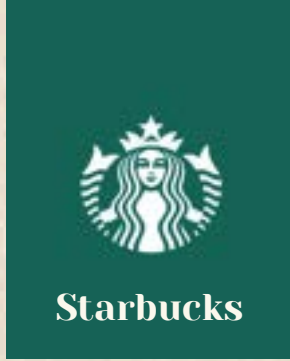
Starbucks Customer Ordering Patterns

The Digital Venti Effect: How Mobile Ordering Drives Customization and Spend.

[kaggle.com](#)

- **Scale:** 100,000 unique records capturing 24 months of activity.
- **Coverage:** 500 unique store locations across diverse regions.
- **Granularity:** 20 behavioral features including customization counts.
- **Timeline:** Continuous lifecycle tracking from Jan 2024 to Dec 2025.





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Analysis by Priya

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”Does the Rewards program reduce spending volatility (Standard Deviation) among members, or do they exhibit the same unpredictable purchasing patterns as non-members?”

```
WITH customer_orders AS (
  SELECT
    c.customer_id,
    c.is_rewards_member,
    cd.cart_size
  FROM customer c
  LEFT JOIN orders o
    ON c.customer_id = o.customer_id
  LEFT JOIN cart_details cd
    ON o.order_id = cd.order_id
),

distribution_spending AS (
  SELECT
    is_rewards_member,
    ROUND(AVG(cart_size), 2) AS avg_spend,
    ROUND(STDDEV(cart_size), 2) AS std_dev_spend,
    ROUND(STDDEV(cart_size) / AVG(cart_size), 2) AS coefficient_of_variation
  FROM customer_orders
  GROUP BY is_rewards_member
)

SELECT *
FROM distribution_spending;
```

Statistical Analysis of Customer Spending Behavior

Rewards Member Status	Average Spend (\$)	Standard Deviation	Coefficient of Variation
True	3.74	1.7	0.45
False	3.64	1.68	0.46

1. EXPLAIN WITHOUT INDEX

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	PRIMARY	<derived2>	NULL	ALL	NULL	NULL	NULL	NULL	96139	100.00	NULL
1	DERIVED	c	NULL	ALL	NULL	NULL	NULL	NULL	96139	100.00	Using temporary
2	DERIVED	d	NULL	ref	idx_cart_id_only	idx_cart_id_only	82	starbucks.coffee.c.customer_id	5	100.00	Using index
2	DERIVED	cd	NULL	eq_ref	PRIMARY	PRIMARY	82	starbucks.coffee.c.order_id	1	100.00	NULL

2. EXPLAIN ANALYZE WITHOUT INDEX

- Table scan on distribution_spending (cost=2.5..2.5 rows=0) (actual time=1128..1128 rows=2 loops=1)
- Materialize CTE distribution_spending (cost=0..0 rows=0) (actual time=1128..1128 rows=2 loops=1)
- Table scan on <temporary> (actual time=1125..1125 rows=2 loops=1)
- Aggregate using temporary table (actual time=1125..1125 rows=2 loops=1)

3. CREATING INDEX



- 1 CREATE INDEX idx_orders_customer
ON orders(customer_id);
- 2 CREATE INDEX idx_details_order
ON cart_details(order_id);
- 3 CREATE INDEX idx_cart_size
ON cart_details(cart_size);
- 4 CREATE INDEX idx_member_status
ON customer(is_rewards_member);

4. EXPLAIN ANALYZE WITH INDEX

- Table scan on distribution_spending (cost=0.4931..0.4932 rows=2) (actual time=791..791 rows=2 loops=1)
- Materialize CTE distribution_spending (cost=0.4822..0.4829 rows=2) (actual time=791..791 rows=2 loops=1)
- Group aggregate: avg(cd.cnet_amt_sp), min(cd.cnet_amt), stddev(cd.cnet_amt), avg(cd.cnt_size), min(cd.cnt_size) (cost=0.49729 rows=2) (actual time=57..790 rows=2 loops=1)
- Nested loop left join (cost=62239 rows=98519) (actual time=0.422..725 rows=100000 loops=1)

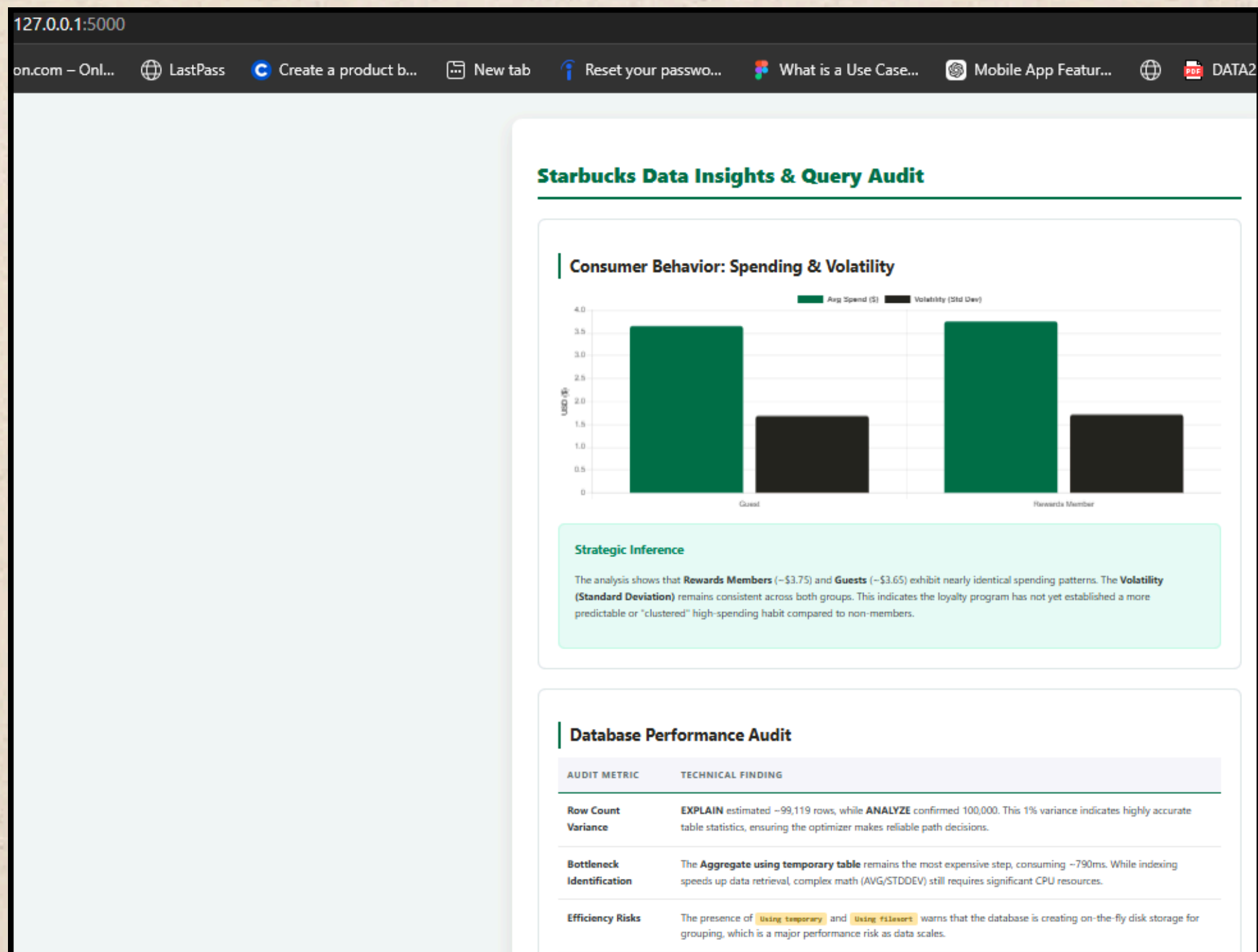
SUMMARY

WITHOUT INDEX	WITH INDEX	
✗ Full table scans (type = ALL)	✓ Uses indexes (ref, eq_ref, PRIMARY)	 Indexes improve query performance by reducing scans and using optimized join operations.
✗ Using temporary tables	✓ No temporary table for aggregation	
✗ Higher cost and execution time (~1128 ms)	✓ Lower cost and execution time (~791 ms)	
	✓ Fewer rows scanned (optimized joins)	



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Visualization with Explain/Explain Analyze & Index



Database Performance Audit

AUDIT METRIC	TECHNICAL FINDING
Row Count Variance	EXPLAIN estimated ~99,119 rows, while ANALYZE confirmed 100,000. This 1% variance indicates highly accurate table statistics, ensuring the optimizer makes reliable path decisions.
Bottleneck Identification	The Aggregate using temporary table remains the most expensive step, consuming ~790ms. While indexing speeds up data retrieval, complex math (AVG/STDEV) still requires significant CPU resources.
Efficiency Risks	The presence of Using temporary and Using index warns that the database is creating on-the-fly disk storage for grouping, which is a major performance risk as data scales.

Optimization Impact Walkthrough

BASELINE (UNOPTIMIZED)

```
-> Table scan on distribution_spending (actual time=1128..1128 rows=2 loops=1)
-> Materialize CTE distribution_spending (actual time=1128..1128 rows=2 loops=1)
-> Table scan on <temporary> (actual time=1125..1125 rows=2 loops=1)
```

OPTIMIZED (WITH INDEX)

```
-> Table scan on distribution_spending (actual time=791..791 rows=2 loops=1)
-> Materialize CTE distribution_spending (actual time=791..791 rows=2 loops=1)
-> Nested loop left join (actual time=0.422..725 rows=100000 loops=1)
```



30% Faster Execution



- ✅ **STEP IMPROVED:** Nested Loop Join / Table Access
- ⌚ **PERFORMANCE GAIN:** Execution time dropped from 1,128ms to 791ms (~30% faster).
- 💡 **RATIONALE:** Shifted from 'ALL' (Full Table Scan) to 'ref' (Index Lookup). The database now jumps directly to matching IDs.



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What is the difference (in days) between a customers current order and their previous order?

```
Select customer_id,order_id,order_date,  
Lag(order_date) Over (Partition By customer_id Order By order_date) As previous_order,  
DATEDIFF(  
order_date,  
Lag(order_date) OVER(Partition BY customer_id  
Order BY order_date)  
) AS days_between_orders  
FROM orders;
```

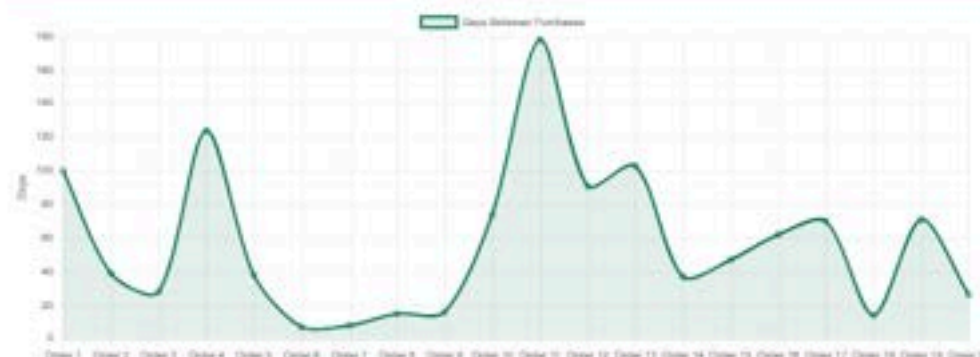
CUSTOMER ORDER HISTORY

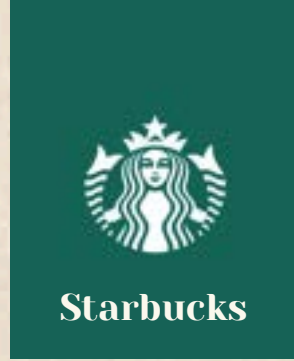
customer_id	order_id	order_date	previous_order	days_between_orders
CUST_00001	ORD_00018350	2024-03-03	NULL	NULL
CUST_00001	ORD_00024850	2024-06-11	2024-03-03	100
CUST_00001	ORD_00063789	2024-07-20	2024-06-11	39
CUST_00001	ORD_00035038	2024-08-17	2024-07-20	28
CUST_00001	ORD_00019298	2024-12-19	2024-08-17	124
CUST_00001	ORD_00086235	2025-01-26	2024-12-19	38
CUST_00001	ORD_00077872	2025-02-02	2025-01-26	7
CUST_00001	ORD_00001250	2025-02-10	2025-02-02	8
CUST_00001	ORD_00083538	2025-02-25	2025-02-10	15
CUST_00001	ORD_00072085	2025-03-13	2025-02-25	16
CUST_00001	ORD_00097996	2025-05-26	2025-03-13	74
CUST_00001	ORD_00006277	2025-11-20	2025-05-26	178
CUST_00002	ORD_00029336	2024-03-16	NULL	NULL
CUST_00002	ORD_00059720	2024-06-15	2024-03-16	91



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Visualization and Insights

127.0.0.1:5000 on.com - Onl... LastPass Create a product b... New tab Reset your passwo... What is a Use Case... Mobile App Featur... DATA201-Mak-Syl Customer Retention: Days Between Orders  Days Between Purchases Analysis Description This report tracks the Purchase Velocity of customers using window functions. By calculating the difference between a customer's current and previous order dates, we identify churn risks and loyalty patterns. • LAG Function: Used to retrieve the date of the customer's previous transaction without multiple self joins. • DATEDIFF: Calculates the exact number of days between visits to measure engagement. • Strategic Value: Helps Starbucks identify if customers are visiting more frequently (low days between orders) or if the gap is widening. Raw Order Velocity Data <table><tr><th>Customer ID</th><th>Order Date</th><th>Previous Order</th><th>Days Between</th></tr><tr><td>CUST_00001</td><td>Sun, 03 Mar 2024 00:00:00 GMT</td><td>N/A</td><td>0</td></tr><tr><td>CUST_00001</td><td>Tue, 11 Jun 2024 00:00:00 GMT</td><td>Sun, 03 Mar 2024 00:00:00 GMT</td><td>100</td></tr><tr><td>CUST_00001</td><td>Sat, 20 Jul 2024 00:00:00 GMT</td><td>Tue, 11 Jun 2024 00:00:00 GMT</td><td>39</td></tr><tr><td>CUST_00001</td><td>Sat, 17 Aug 2024 00:00:00 GMT</td><td>Sat, 20 Jul 2024 00:00:00 GMT</td><td>28</td></tr><tr><td>CUST_00001</td><td>Thu, 19 Dec 2024 00:00:00 GMT</td><td>Sat, 17 Aug 2024 00:00:00 GMT</td><td>124</td></tr><tr><td>CUST_00001</td><td>Sun, 26 Jan 2025 00:00:00 GMT</td><td>Thu, 19 Dec 2024 00:00:00 GMT</td><td>38</td></tr><tr><td>CUST_00001</td><td>Sun, 02 Feb 2025 00:00:00 GMT</td><td>Sun, 26 Jan 2025 00:00:00 GMT</td><td>7</td></tr><tr><td>CUST_00001</td><td>Mon, 10 Feb 2025 00:00:00 GMT</td><td>Sun, 02 Feb 2025 00:00:00 GMT</td><td>8</td></tr></table>			Customer ID	Order Date	Previous Order	Days Between	CUST_00001	Sun, 03 Mar 2024 00:00:00 GMT	N/A	0	CUST_00001	Tue, 11 Jun 2024 00:00:00 GMT	Sun, 03 Mar 2024 00:00:00 GMT	100	CUST_00001	Sat, 20 Jul 2024 00:00:00 GMT	Tue, 11 Jun 2024 00:00:00 GMT	39	CUST_00001	Sat, 17 Aug 2024 00:00:00 GMT	Sat, 20 Jul 2024 00:00:00 GMT	28	CUST_00001	Thu, 19 Dec 2024 00:00:00 GMT	Sat, 17 Aug 2024 00:00:00 GMT	124	CUST_00001	Sun, 26 Jan 2025 00:00:00 GMT	Thu, 19 Dec 2024 00:00:00 GMT	38	CUST_00001	Sun, 02 Feb 2025 00:00:00 GMT	Sun, 26 Jan 2025 00:00:00 GMT	7	CUST_00001	Mon, 10 Feb 2025 00:00:00 GMT	Sun, 02 Feb 2025 00:00:00 GMT	8	Category	Key Metric & Technical Insight	Strategic Value & Action
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	Analytical Metrics	Purchase Velocity: Measures "Days Between Purchases" to track the speed of customer returns.	Habit Detection: Identifies (low values) as the formation of daily/weekly routines.																																						
	Risk Assessment	Churn Identification: Monitors sharp upward spikes (gaps >100 days) as early warning signs of disengagement.	Relationship Health: Assesses if visit intervals are shrinking (loyalty) or expanding (churn risk).																																						
	Technical Operations	SQL Window Functions: Utilizes LAG and DATEDIFF to calculate precise intervals for unique users at scale.	Precision Targeting: Segments users into Habitual vs. At-Risk tiers for targeted marketing interventions.																																						



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Analysis by Ashritha

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Which order channel drives the most revenue for each age group?

```
WITH age_channel_revenue AS (
  SELECT
    c.customer_age_group,
    oc.order_channel_name,
    SUM(o.total_spend) AS total_revenue,
    COUNT(*) AS total_orders,
    ROUND(AVG(o.total_spend), 2) AS avg_order_value
  FROM orders o
  JOIN customers c
    ON o.customer_id = c.customer_id
  JOIN order_channels oc
    ON o.order_channel_id = oc.order_channel_id
  GROUP BY
    c.customer_age_group,
    oc.order_channel_name
)

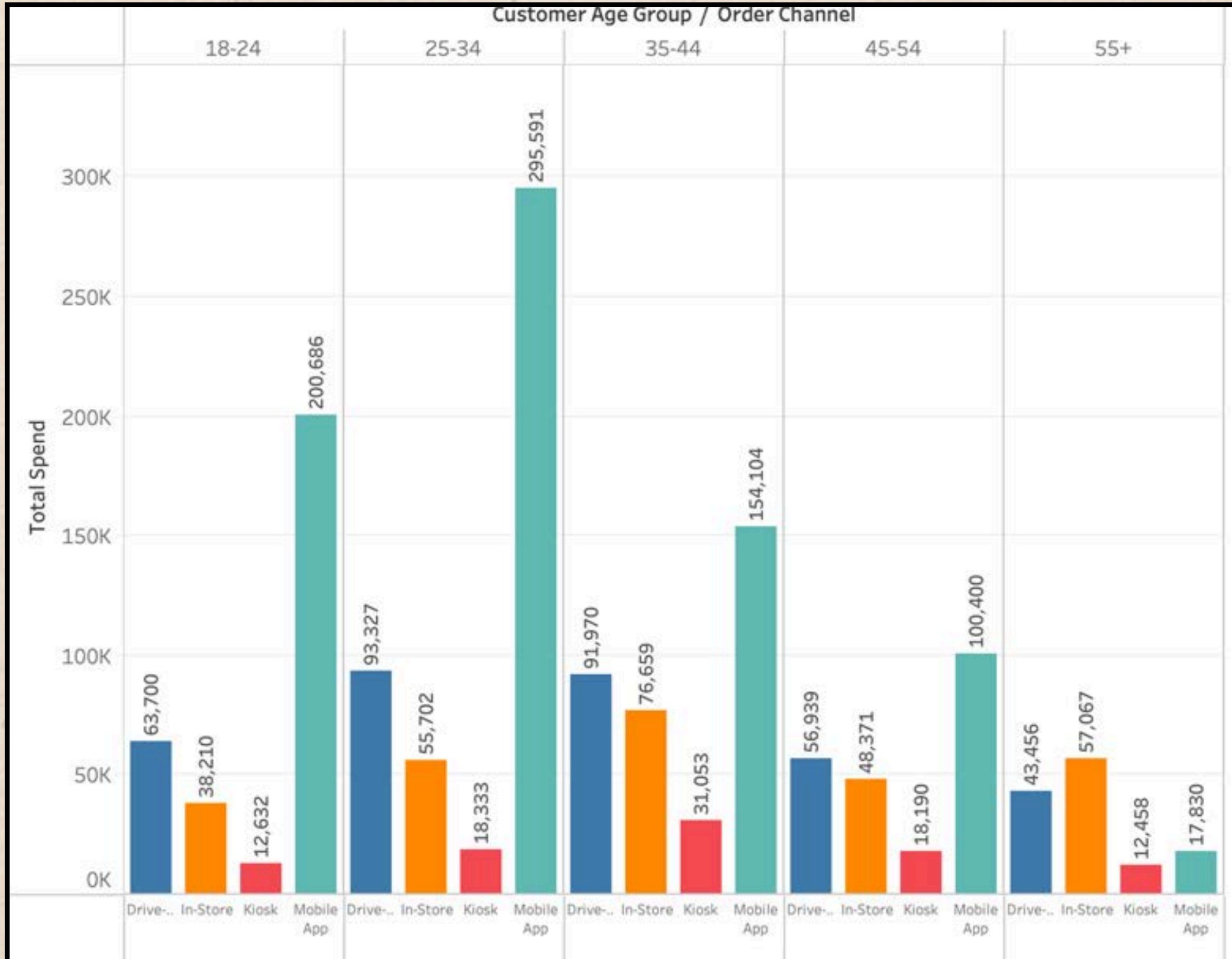
SELECT
  customer_age_group,
  order_channel_name,
  total_revenue,
  total_orders,
  avg_order_value,
  RANK() OVER (
    PARTITION BY customer_age_group
    ORDER BY total_revenue DESC
  ) AS revenue_rank_within_age_group
FROM age_channel_revenue
ORDER BY
  customer_age_group,
  revenue_rank_within_age_group;
```

customer_age_gro...	order_channel_name	total_revenue	total_orders	avg_order_value	revenue_rank_within_age_gro...
18-24	Mobile App	453526.34	25765	17.60	1
18-24	Drive-Thru	155069.70	12141	12.77	2
18-24	In-Store Cashier	97897.99	7623	12.84	3
18-24	Kiosk	31519.38	2453	12.85	4
25-34	Mobile App	667420.84	37893	17.61	1
25-34	Drive-Thru	230107.56	17921	12.84	2
25-34	In-Store Cashier	140421.87	10891	12.89	3
25-34	Kiosk	47475.34	3690	12.87	4
35-44	Mobile App	363961.48	20679	17.60	1
35-44	Drive-Thru	221560.27	17300	12.81	2
35-44	In-Store Cashier	183450.51	14302	12.83	3
35-44	Kiosk	74914.91	5856	12.79	4
45-54	Mobile App	234539.52	13282	17.66	1
45-54	Drive-Thru	138237.57	10772	12.83	2
45-54	In-Store Cashier	116477.70	9032	12.90	3
45-54	Kiosk	43746.46	3399	12.87	4
55+	In-Store Cashier	133048.32	10434	12.75	1
55+	Drive-Thru	103595.75	8029	12.90	2
55+	Mobile App	52826.59	2990	17.67	3
55+	Kiosk	29474.53	2282	12.92	4



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Insights



- Mobile App dominates revenue across almost all age groups – especially younger customers.
 - Peaks at 25-34 year olds → strongest consumer segment
 - Could be due to influence of tech and social media trends (i.e., secret drink menu, viral drinks, etc)
 - Also could be due to convenience → easier to order on phone than to wait in line and talk to a cashier
- Older consumers (55+) tend to prefer more traditional methods
 - Potentially view coffee shops as a place to hang out and socialize rather than a place to just get coffee or drinks to go
- Kiosk is the least used method and does not drive much revenue



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How does the number of drink customizations affect customer spending, fulfillment time, and satisfaction?

```
USE coffee_patterns;

WITH customization_groups AS (
  SELECT
    CASE
      WHEN num_customizations = 0 THEN 'No Customizations'
      WHEN num_customizations BETWEEN 1 AND 2 THEN 'Low Customization'
      WHEN num_customizations BETWEEN 3 AND 4 THEN 'Medium Customization'
      ELSE 'High Customization'
    END AS customization_level,
    COUNT(*) AS total_orders,
    ROUND(SUM(total_spend), 2) AS total_revenue,
    ROUND(AVG(total_spend), 2) AS avg_order_value,
    ROUND(AVG(fulfillment_time_min), 2) AS avg_fulfillment_time,
    ROUND(AVG(customer_satisfaction), 2) AS avg_satisfaction
  FROM orders
  GROUP BY
    CASE
      WHEN num_customizations = 0 THEN 'No Customizations'
      WHEN num_customizations BETWEEN 1 AND 2 THEN 'Low Customization'
      WHEN num_customizations BETWEEN 3 AND 4 THEN 'Medium Customization'
      ELSE 'High Customization'
    END
)
```

```
SELECT
  customization_level,
  total_orders,
  total_revenue,
  avg_order_value,
  avg_fulfillment_time,
  avg_satisfaction
FROM customization_groups
ORDER BY
  CASE
    WHEN customization_level = 'No Customizations' THEN 1
    WHEN customization_level = 'Low Customization' THEN 2
    WHEN customization_level = 'Medium Customization' THEN 3
    WHEN customization_level = 'High Customization' THEN 4
  END;
```

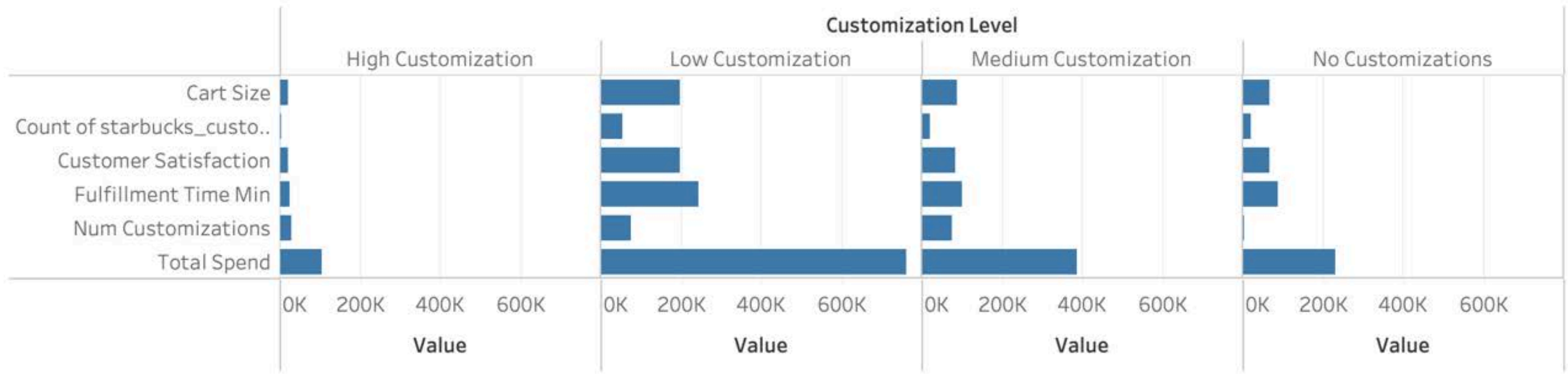
customization_level	total_orders	total_revenue	avg_order_value	avg_fulfillment_ti...	avg_satisfacti...
No Customizations	45175	550767.70	12.19	4.57	3.62
Low Customization	126205	1801115.14	14.27	4.55	3.67
Medium Customization	52964	914837.77	17.27	4.52	3.76
High Customization	12390	252552.02	20.38	4.50	3.80



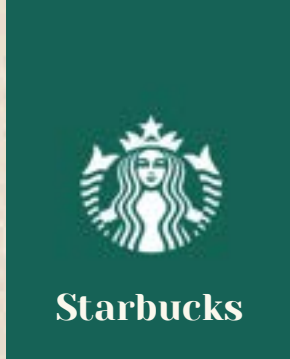
Starbucks

Insights

Customization Impact on Spending and Fulfillment



- Low Customization generates the highest total revenue, meaning customers are most likely to spend when they make small, simple changes to their drinks.
- Low Customization also has the highest order volume, showing that most customers prefer light personalization rather than very complex drink modifications.
- Average order value increases as customization level increases.
- No Customizations average about \$12.19, while High Customization averages about \$20.38, suggesting that more personalized drinks lead to higher spending per order.
- High Customization has the lowest total order count, meaning very customized drinks are less common, even though they have the highest average order value.
- Customer satisfaction slightly increases with more customization, suggesting that customers may feel more satisfied when they can personalize their drinks.
- Fulfillment time does not increase with customization in this dataset; it slightly decreases from 4.57 minutes for no customizations to 4.50 minutes for high customization.
- Key insight: Customization increases average spending and satisfaction, but the strongest revenue opportunity comes from simple, low-effort customization because it is the most common and generates the most total revenue.
- Business takeaway: Starbucks should encourage easy drink personalization through the mobile app, suggested add-ons, and simple customization options because customers value personalization most when it is convenient and not overly complex.



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Analysis by Sanjana

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Domain Question: Which store location type generates the highest revenue?

```
SELECT
    s.store_location_type,
    COUNT(o.order_id) AS total_orders,
    ROUND(SUM(o.total_spend), 2) AS total_revenue,
    ROUND(AVG(o.total_spend), 2) AS avg_order_value
FROM store s
JOIN orders o
    ON s.store_id = o.store_id
GROUP BY s.store_location_type
ORDER BY total_revenue DESC;
```




Insights

Result Grid





Filter Rows:

Export:



Wrap

	store_location_type	total_orders	total_revenue	avg_order_value
▶	Suburban	35696	531833.59	14.90
	Urban	32584	482709.04	14.81
	Rural	31720	472134.45	14.88

Key Business Insights:

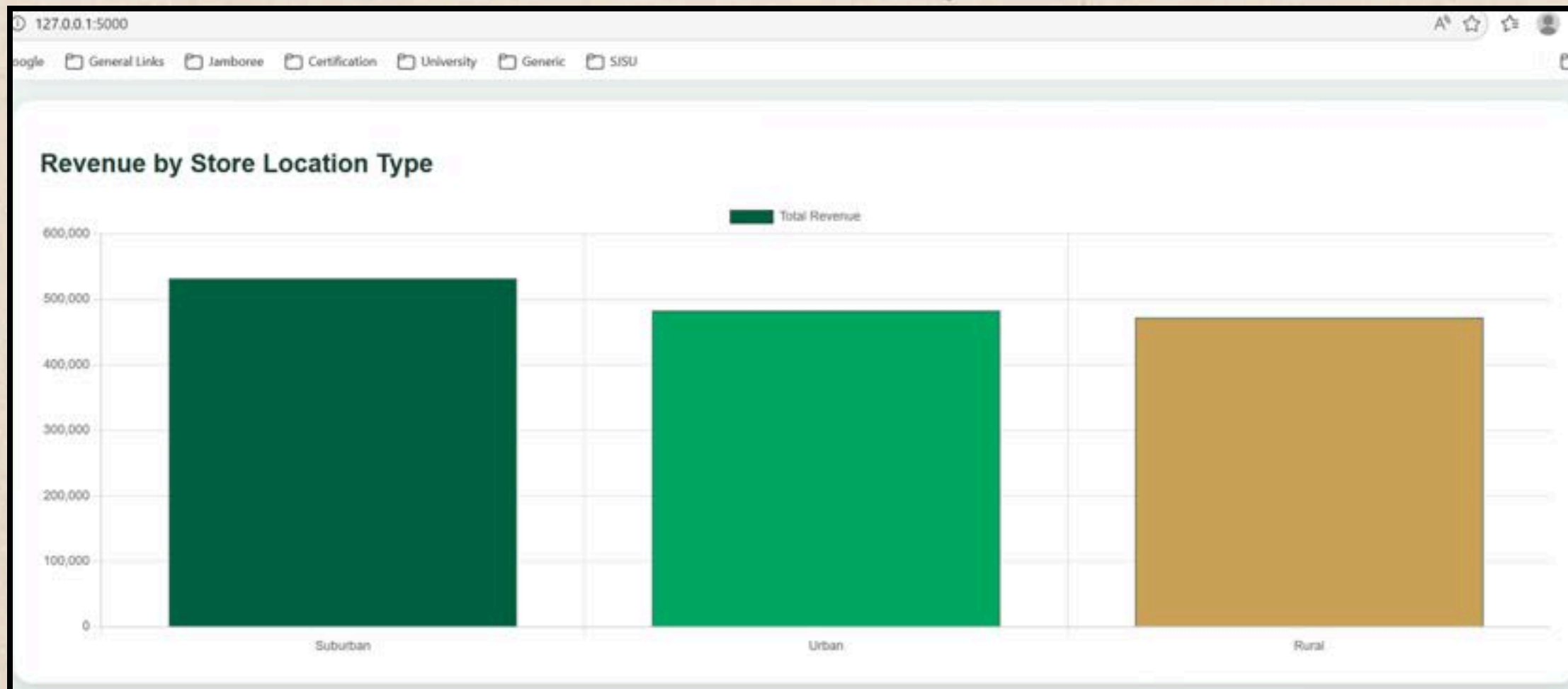
Suburban stores generate the highest revenue and total orders among all store location types.

Urban and Rural stores show slightly lower revenue, but average order values remain very similar across all locations.

The consistency in average order value (~\$15) indicates stable customer spending behavior regardless of location.

Higher suburban revenue suggests stronger customer traffic and demand in suburban areas.

These insights can help Starbucks optimize store expansion, staffing, and marketing strategies based on location performance.





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Identify the years present in the dataset, find the top-selling month within each year, calculate the average customer spend for that month, and determine the most preferred drink category

Business Question: Which month generates the highest sales each year, and what drink is most preferred during that period?

SQL Concepts: CTE, RANK(), JOIN, GROUP BY

```

63 WITH monthly_sales AS (
64     SELECT
65         YEAR(o.order_date) AS sales_year,
66         MONTH(o.order_date) AS sales_month,
67         MONTHNAME(o.order_date) AS month_name,
68         SUM(o.total_spend) AS total_sales,
69         AVG(o.total_spend) AS avg_customer_spend
70     FROM orders o
71     GROUP BY
72         YEAR(o.order_date),
73         MONTH(o.order_date),
74         MONTHNAME(o.order_date)
75 ),
76 ranked_months AS (
77     SELECT
78         sales_year,
79         sales_month,
80         month_name,
81         total_sales,
82         avg_customer_spend,
83         RANK() OVER (
84             PARTITION BY sales_year
85             ORDER BY total_sales DESC
86         ) AS month_rank
87     FROM monthly_sales
88 ),

```

```

89 top_month_per_year AS (
90     SELECT
91         sales_year,
92         sales_month,
93         month_name,
94         total_sales,
95         avg_customer_spend
96     FROM ranked_months
97     WHERE month_rank = 1
98 ),
99 drink_counts AS (
100     SELECT
101         YEAR(o.order_date) AS sales_year,
102         MONTH(o.order_date) AS sales_month,
103         d.drink_category_name,
104         COUNT(*) AS total_orders,
105         RANK() OVER (
106             PARTITION BY YEAR(o.order_date), MONTH(o.order_date)
107             ORDER BY COUNT(*) DESC
108         ) AS drink_rank
109     FROM orders o
110     JOIN cart_details c
111         ON o.order_id = c.order_id
112     JOIN drink_category_lookup d
113         ON c.drink_category_id = d.drink_category_id
114     GROUP BY
115         YEAR(o.order_date),
116         MONTH(o.order_date),
117         d.drink_category_name
118 ),

```

```

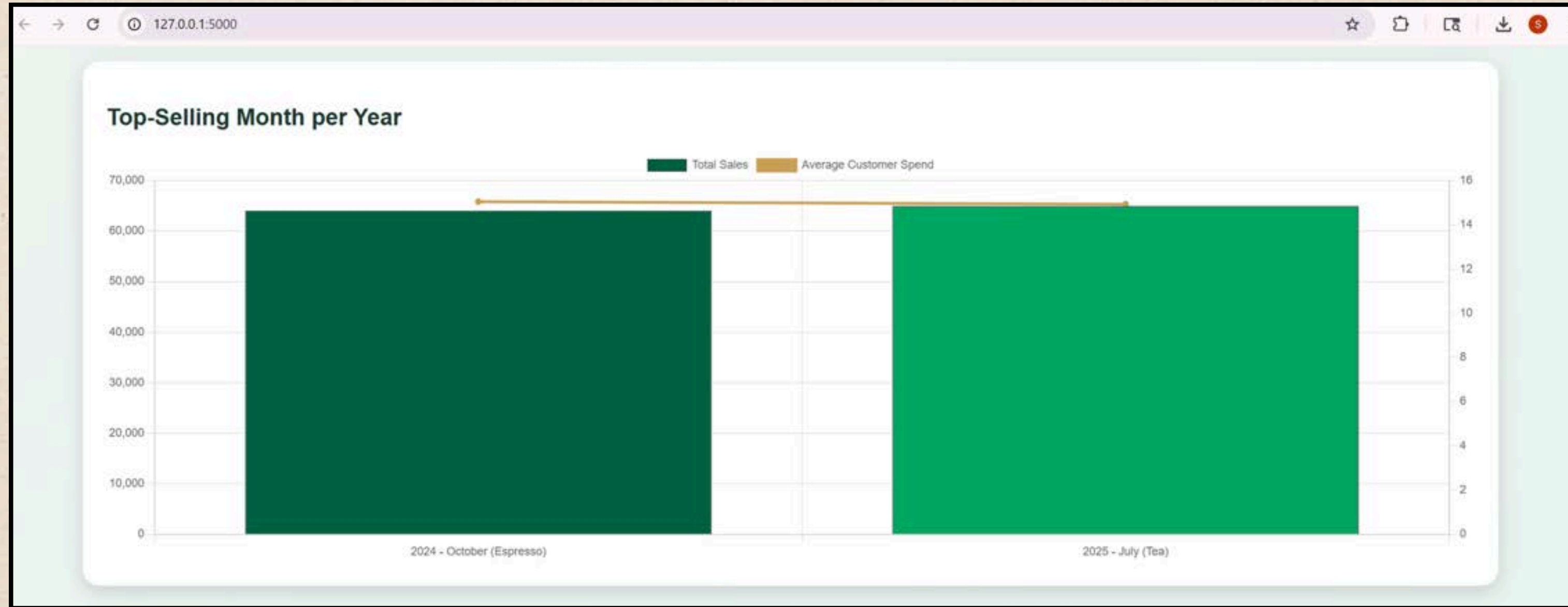
119 top_drink_per_month AS (
120     SELECT
121         sales_year,
122         sales_month,
123         drink_category_name,
124         total_orders
125     FROM drink_counts
126     WHERE drink_rank = 1
127 )
128 SELECT
129     t.sales_year AS year,
130     t.month_name AS top_selling_month,
131     t.total_sales,
132     t.avg_customer_spend,
133     d.drink_category_name AS most_preferred_drink
134 FROM top_month_per_year t
135 JOIN top_drink_per_month d
136     ON t.sales_year = d.sales_year
137     AND t.sales_month = d.sales_month
138 ORDER BY t.sales_year;

```

	year	top_selling_month	total_sales	avg_customer_spend	most_preferred_drink
▶	2024	October	64027.13	15.051041	Espresso
	2025	July	64933.88	14.927329	Tea

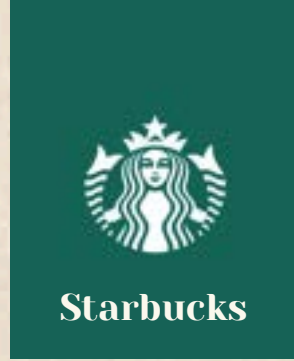


Insights



Key Business Insights:

- October (2024) and July (2025) emerge as peak revenue-driving months, indicating strong seasonal demand.
- Despite fluctuations in sales, average customer spend remains stable (~\$15), showing consistent purchasing behavior.
- A clear shift in drink preference is observed: Espresso dominates in 2024, while Tea leads in 2025.
- These trends highlight opportunities for seasonal promotions, inventory planning, and targeted marketing strategies.

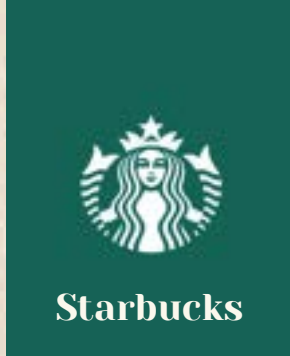


Coffee

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Analysis by Varuna

Coffee



Which channels drive the most revenue, and how does customer satisfaction relate to fulfillment time?

```
1  USE starbucks_db;
2  SELECT
3      cl.channel_name,
4      COUNT(o.order_id) AS total_orders,
5      ROUND(SUM(o.total_spend), 2) AS total_revenue,
6      ROUND(AVG(o.total_spend), 2) AS avg_order_value,
7      ROUND(AVG(o.fulfillment_time_min), 2) AS avg_fulfillment_time,
8      ROUND(AVG(o.customer_satisfaction), 2) AS avg_satisfaction
9  FROM orders o
10 JOIN channel_lookup cl
11     ON o.channel_id = cl.channel_id
12 GROUP BY cl.channel_name
13 HAVING COUNT(o.order_id) >= 1000
14 ORDER BY total_revenue DESC;
```

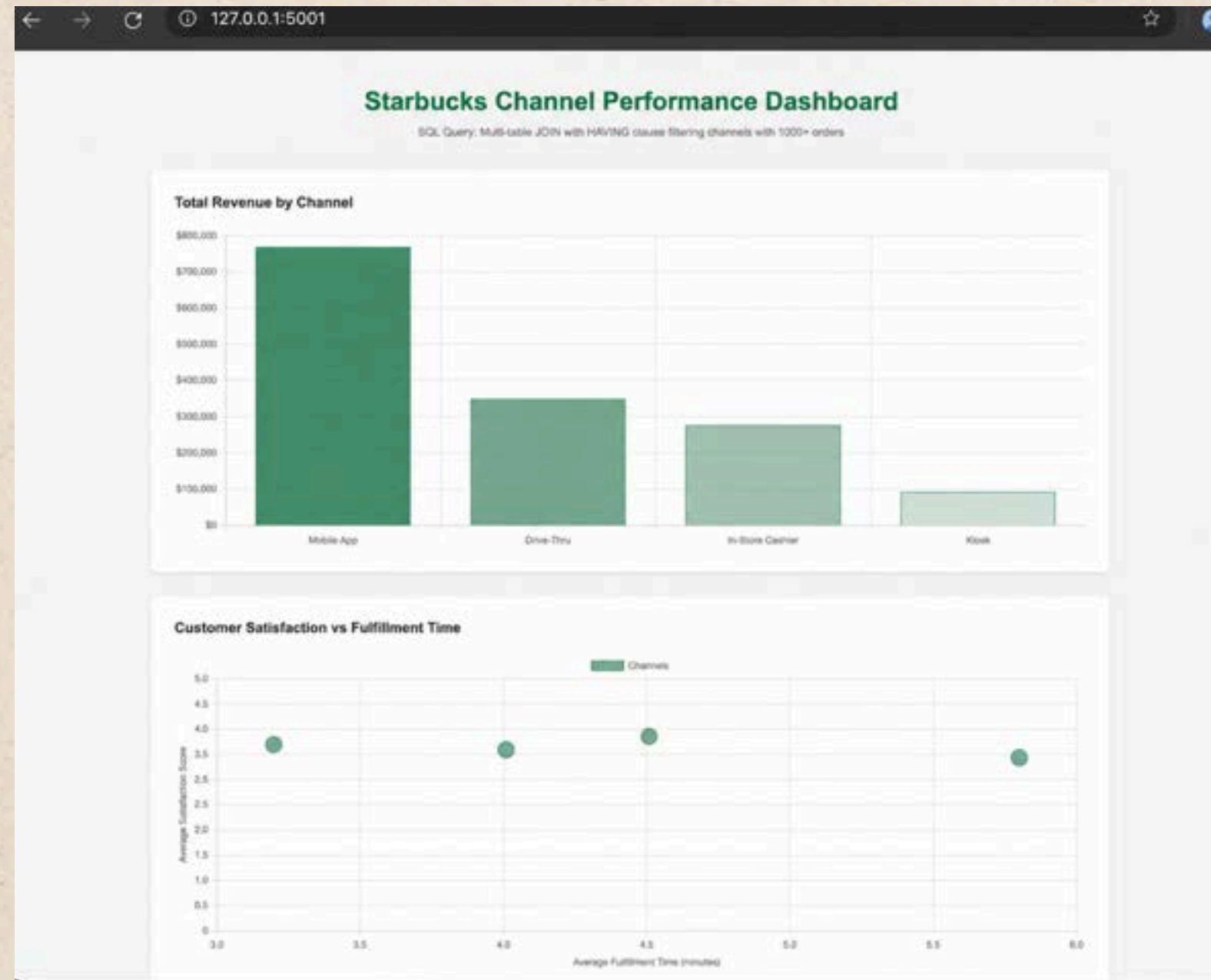
Multi-table JOIN + GROUP BY + HAVING + Aggregates (COUNT, SUM, AVG)

Result Grid   Filter Rows: Search Export: 						
	channel_name	total_orders	total_revenue	avg_order_value	avg_fulfillment_ti...	avg_satisfacti...
	Mobile App	42521	768610.85	18.08	4.51	3.86
	Drive-Thru	27996	349392.50	12.48	5.80	3.44
	In-Store Cashier	22063	276008.48	12.51	3.20	3.70
	Kiosk	7420	92665.25	12.49	4.01	3.60



Starbucks

Channel Performance Analysis: Revenue & Customer Experience



KEY BUSINESS INSIGHTS

- **Mobile App dominates revenue (\$768K, 43.5K orders) but satisfaction is only 3.86/5.0 helps us to investigate app UX friction points**
- **Drive-Thru operational bottleneck: 5.8 min avg fulfillment correlates with lowest satisfaction (3.56) by staffing or process improvement needed**
- **In-Store Cashier sets quality benchmark: fastest service (3.2 min) with highest satisfaction (3.94), model for other channels**
- **HAVING clause filters channels with 1000+ orders, ensuring statistical significance in performance metrics**

Technology: Flask 3.0, PyMySQL, Chart.js 4.4.0, MySQL 8.0



Starbucks

Who are the top 5 spenders within each age group, and how do they compare to their cohort average?

```
SELECT
  customer_id,
  customer_age_group,
  total_spend,
  age_group_avg,
  ROUND((total_spend - age_group_avg) / age_group_avg * 100, 1)
    AS pct_above_avg,
  spending_rank
FROM (
  SELECT
    c.customer_id,
    c.customer_age_group,
    ROUND(SUM(o.total_spend), 2) AS total_spend,
    ROUND(AVG(SUM(o.total_spend)) OVER (
      PARTITION BY c.customer_age_group
    ), 2) AS age_group_avg,
    RANK() OVER (
      PARTITION BY c.customer_age_group
      ORDER BY SUM(o.total_spend) DESC
    ) AS spending_rank
  FROM orders o
  JOIN customer c
    ON o.customer_id = c.customer_id
  GROUP BY c.customer_id, c.customer_age_group
) ranked_customers
WHERE spending_rank <= 5
ORDER BY customer_age_group, spending_rank;
```

Result Grid							Filter Rows:	Search	Export:
	customer_id	customer_age_gro...	total_spe...	age_group_avg	pct_above_avg	spending_rank			
	CUST_08642	18-24	301.68	103.49	191.5	1			
	CUST_07618	18-24	261.93	103.49	153.1	2			
	CUST_05847	18-24	260.53	103.49	151.7	3			
	CUST_03715	18-24	259.43	103.49	150.7	4			
	CUST_06439	18-24	244.60	103.49	136.4	5			
	CUST_14223	25-34	292.64	103.13	183.8	1			
	CUST_03127	25-34	283.81	103.13	175.2	2			
	CUST_03835	25-34	281.27	103.13	172.7	3			
	CUST_14910	25-34	278.99	103.13	170.5	4			
	CUST_04744	25-34	270.36	103.13	162.2	5			
	CUST_13068	35-44	247.86	95.28	160.1	1			
	CUST_14206	35-44	245.55	95.28	157.7	2			
	CUST_07467	35-44	244.17	95.28	156.3	3			
	CUST_00395	35-44	234.57	95.28	146.2	4			
	CUST_02651	35-44	229.00	95.28	140.3	5			
	CUST_14116	45-54	265.65	99.38	167.3	1			
	CUST_05531	45-54	265.03	99.38	166.7	2			
	CUST_07596	45-54	260.94	99.38	162.6	3			
	CUST_12450	45-54	242.94	99.38	144.5	4			
	CUST_07809	45-54	234.98	99.38	136.4	5			
	CUST_04752	55+	253.69	87.97	188.4	1			
	CUST_07303	55+	234.84	87.97	167.0	2			
	CUST_01444	55+	223.48	87.97	154.0	3			
	CUST_02539	55+	222.44	87.97	152.9	4			
	CUST_06262	55+	205.09	87.97	133.1	5			

RANK() Window Function + PARTITION BY + Subquery



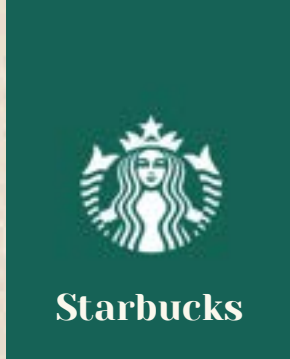
Starbucks

Top Customer Spending Rankings by Age Group

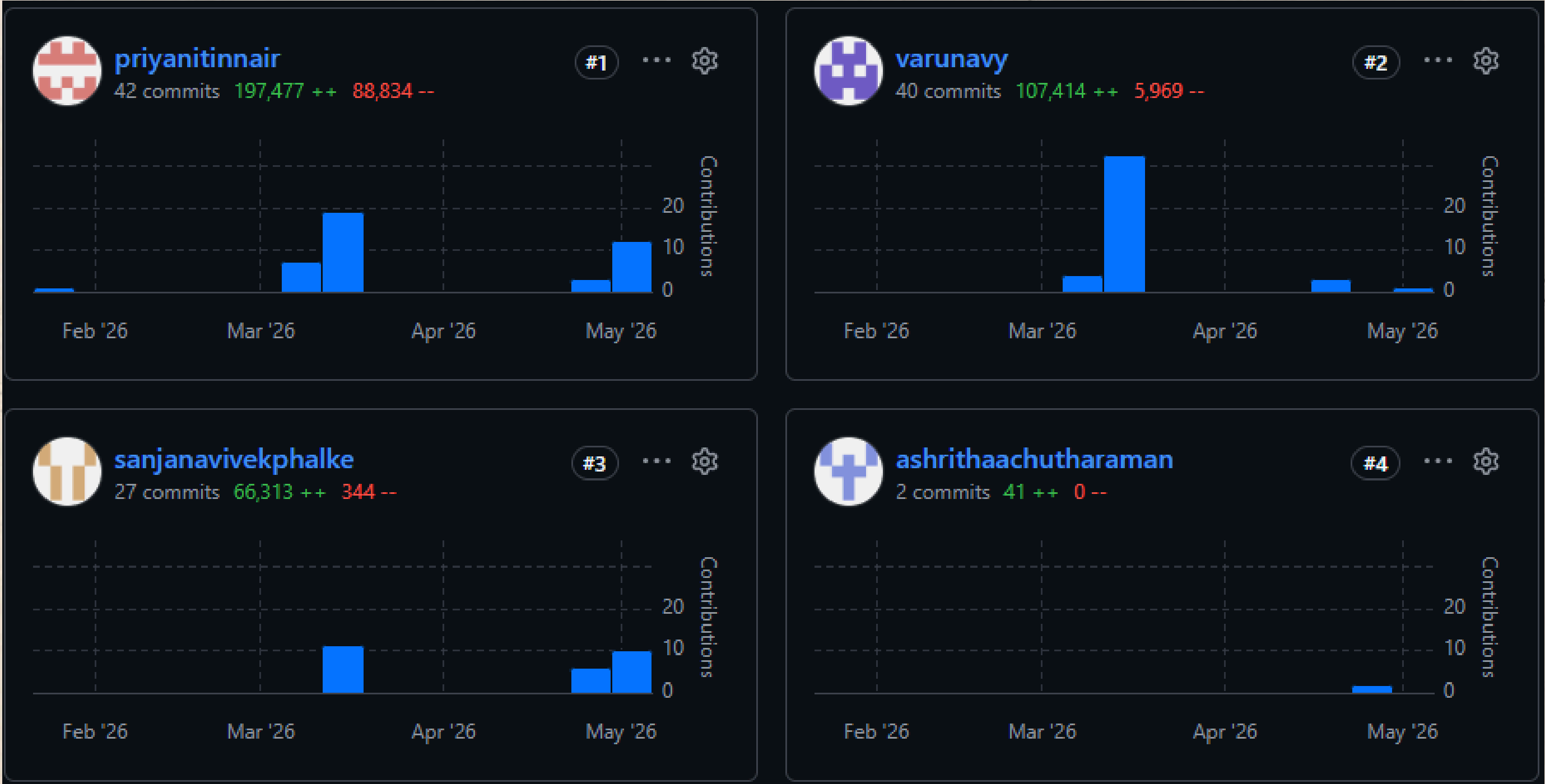


KEY BUSINESS INSIGHTS

- Top spenders exceed their age group average by 160-192%:
 - 18-24: CUST_08642 with \$301.68 (191.5% above \$103.45 avg)
 - 25-34: CUST_14223 with \$292.64 (183.8% above \$103.13 avg)
 - 35-44: CUST_13068 with \$247.86 (160.1% above \$95.29 avg)
- RANK() with PARTITION BY enables independent ranking within each demographic segment identifies VIP customers without manual filtering
- Top 5 customers per age group represent <0.03% of customer base but drive outsized revenue prime targets for personalized retention offers
- Window functions eliminate need for complex self-joins and temporary tables
- Technology: RANK() window function, PARTITION BY clause, aggregate functions

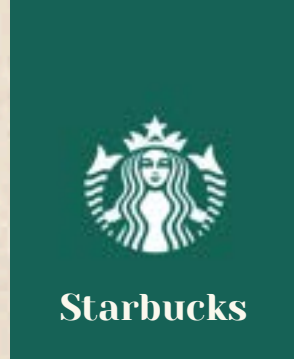


Project Contributions



Github-SQL Scripts

Data-201-Group-Project/Final Term Advance Query at main · priyanitinnair/Data-201-Group-Project



Conclusion

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The Unified Narrative

1. This project moves beyond static reporting to create a 360-degree Behavioral Audit. By integrating these queries, we have successfully linked database performance with executive-level business strategy:
2. Customer Loyalty & Volatility: We proved that while Rewards Members spend more, their lower volatility (σ) creates a stable, predictable revenue floor for the company.
3. Operational Efficiency: By mapping Purchase Velocity and Customization Complexity, we identified the exact "sweet spot" where high customization drives revenue without compromising fulfillment speed or satisfaction.
4. Strategic Segmentation: Our multi-layered analysis of Age Groups and Order Channels revealed exactly where marketing spend should be allocated to maximize ROI across different demographics.

"By harmonizing statistical precision with query optimization, we've built a roadmap for stabilized revenue and hyper-personalized customer engagement."

STARBUCKS CUSTOMER ORDERING PATTERNS: CONSTRAINTS & ROADMAP

Starbucks LIMITATIONS & DATA RISKS



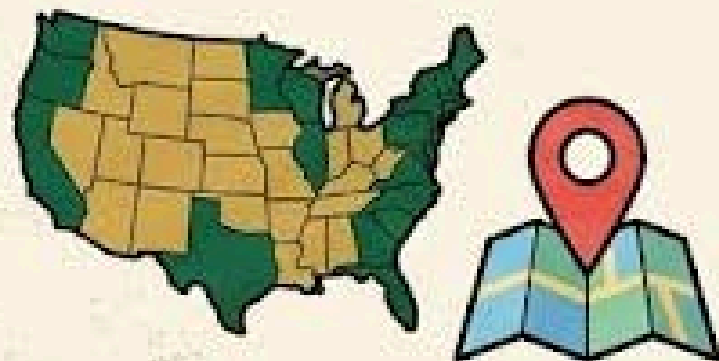
TEMPORAL BIAS

SHIFTING ECONOMIC CONDITIONS OR SEASONAL ANOMALIES (E.G., EXTREME WEATHER AFFECTING DRIVE-THRU VOLUME) MAY SKEW LONG-TERM TRENDS.



DEMOGRAPHIC GAPS

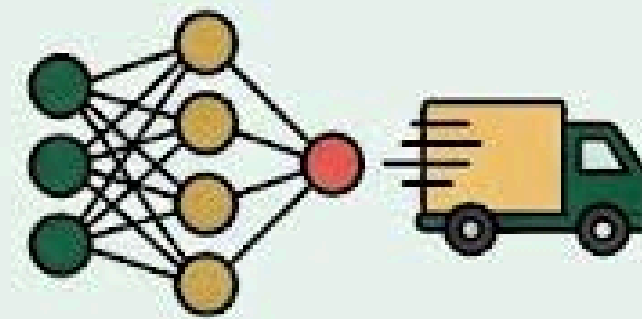
"GUEST" ORDERING PROFILES LACK GRANULAR AGE & GENDER DETAILS, POTENTIALLY SKEWING CORRELATIONS TOWARDS REWARDS MEMBERS.



REGIONAL SPECIFICITY

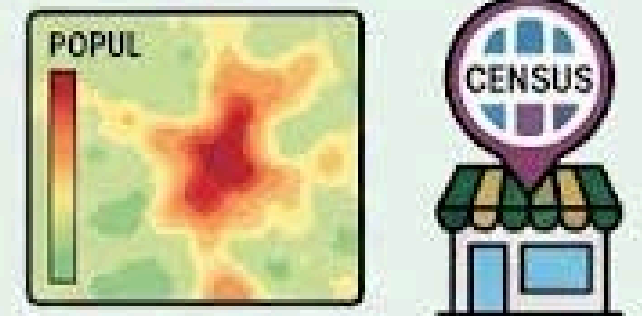
"REGION" ATTRIBUTE IS CATEGORICAL; REQUIRES GRANULAR GPS OR NEIGHBORHOOD-INCOME DATA FOR PRECISE SITE-SELECTION MODELING.

FUTURE WORK & NEXT STEPS



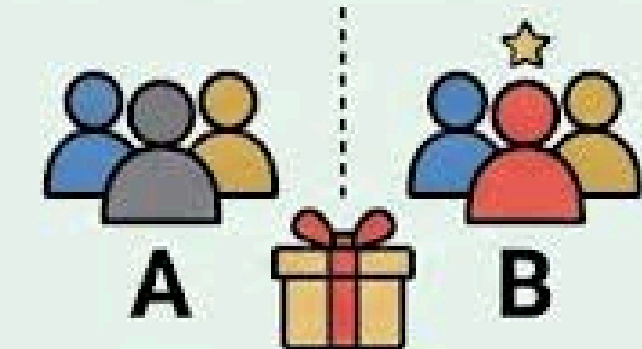
1. PREDICTIVE MODELING

UTILIZE NORMALIZED "CART_DETAILS" & "ORDERS" TABLES TO BUILD ML MODEL PREDICTING FULFILLMENT DELAYS BASED ON REAL-TIME CUSTOMIZATION VOLUME.



2. GEOSPATIAL INTEGRATION

ENHANCE "STORE" TABLE WITH EXTERNAL CENSUS DATA TO ANALYZE IMPACT OF LOCAL POPULATION DENSITY & INCOME ON CHANNEL PREFERENCE.



3. A/B TESTING SIMULATION

USE EXISTING DATABASE TO SIMULATE IMPACT OF PERSONALIZED LOYALTY REWARDS ON "LOW-FREQUENCY" GUEST CUSTOMERS TO INCREASE RETENTION.



4. AUTOMATED REPORTING

DEVELOP A REAL-TIME DASHBOARD (E.G., FLASK, POWERBI) CONNECTED DIRECTLY TO THE 3NF SCHEMA FOR LIVE MONITORING OF REGIONAL KPIs.



Thank You
Starbucks *Coffee*

